State of Crypto

Issue 10 / October 2023

21shares



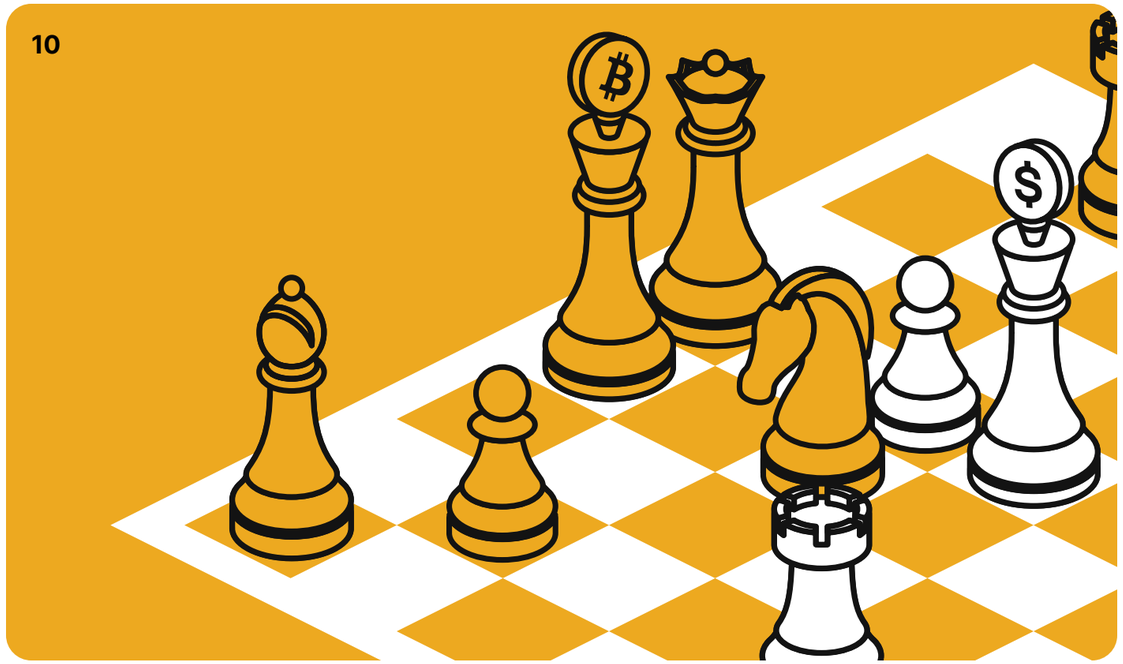
The Top Use Cases

How Crypto is Changing the World

21shares™

The World's First and Largest Crypto ETP Issuer

Digital asset investments for everyone, with simple access to your trades through your broker. Easy, safe and fully regulated.



Contents



10
Bitcoin: the international hedge against wealth confiscation and counterparty risk

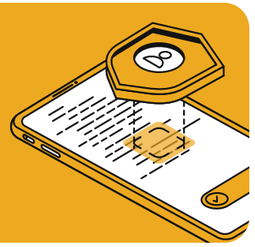
16
USD stablecoins: near-instant and cheap cross-border payments with digital dollars



20
Data storage: an immutable hard drive for preserving humanity's knowledge

26
Digital identity: protecting online authenticity

32
NFTs: democratizing revenue for artists and musicians



38
Forensic science: blockchain-as-a-service to combat illicit activities

42
Conclusion

42
References

43
Disclaimer

Executive Summary

- **Bitcoin, the international hedge against wealth confiscation and counterparty risk:** Bitcoin was created in response to the 2009 Global Financial Crisis as a neutral, international, and censorship-resistant asset and payment infrastructure outside central banks' control. Over the years, people have increasingly adopted Bitcoin as an **emerging store of value**, particularly in countries with high inflation, geopolitical tensions, and economic uncertainty. However, due to the U.S. dollar's status as the global reserve currency, many prefer USD stablecoins or digital dollars over Bitcoin to hedge against a local currency debasement, albeit exposing themselves to counterparty risk.

- **USD stablecoins, near-instant and cheap cross-border payments with digital dollars:** Blockchain networks enable us to transfer value as easily as sending emails globally and almost instantly. We argue that crypto is set to disrupt the **$37 trillion** cross-border payment industry. Where the internet is broadly accessible, people already use stablecoins for remittances. Meanwhile, stablecoin use for merchant payments could grow widespread in the coming years as crypto integrates traditional payment methods like Visa and PayPal.

- **Data storage, an immutable hard drive for preserving humanity's knowledge:** Data storage solutions protect against data loss and censorship. By leveraging the blockchain's distributed architecture, these protocols help preserve humanity's most essential information securely and cheaply, even against surveillance and government censorship. Arweave, for instance, hosts records of Russian violations in Ukraine and censored content of Hong Kong's pro-democracy tabloid, Apple Daily.

- **Digital identity, protecting online authenticity:** Blockchains can protect identification documents (IDs) by storing them permanently and safeguarding them from loss, damage, data breaches, and identity theft. The latter alone cost the world **$8.8 billion** in 2022. Today, decentralized identity protocols account for just 0.07% of the crypto industry's market value, but **over four million people already use the top decentralized IDs**.

- **NFTs, democratizing revenue for artists and musicians:** Traditionally, intermediaries take **88%** of the artists' revenue through commission fees to dealers, galleries, or platforms. With the help of non-fungible tokens (NFTs) and smart contracts, artists and musicians can now retain more than **98%** of the revenue generated by their work, including automatic royalties from secondary market sales. The entry of established brands to Web 3, like Nike, which has registered **$186 million from NFT sales**, illustrates the size of the opportunity.

- **Forensic science, blockchain-as-a-service to combat illicit activities:** Contrary to misconceptions, crypto's usage for illicit activities is minimal. According to Chainalysis, criminal activity constituted **less than 0.3% of crypto's total volume in 2022**. The inherent transparency of blockchain technology enables law enforcement to trace transactions to a degree that isn't possible in traditional finance. Nonetheless, this transparency raises new challenges, including navigating regulatory compliance and addressing privacy concerns.



Introduction

We are thrilled to release our newest State of Crypto magazine - **The Top Use Cases: How Crypto is Changing the World**.

Investors who view crypto solely as a speculative asset class are missing the forest for the trees. Bitcoin and subsequent innovations including smart contracts created a global network for value transfer, similar to how the internet created a global and open network for information. Beyond money, blockchain technology has the potential to reshape the foundations of provenance, data integrity, and ownership in the digital age, especially in an AI-driven world. This new edition provides an overview of how crypto is already being used worldwide, from migrants sending remittances to loved ones abroad and refugees protecting their identity, to redefining how artists receive royalties for their work, and more.

Whether you're a seasoned investor or just getting started, this report is a must-read for anyone interested in this disruptive industry.

Hany Rashwan, CEO
Ophelia Snyder, President
Eliézer Ndinga, Director of Research

"Beyond money, blockchain technology has the potential to reshape the foundations of provenance, data integrity, and ownership in the digital age, especially in an AI-driven world."



About Our Research

21Shares is the world's leader in providing access to crypto through simple and easy-to-use products — co-founded by Hany Rashwan and Ophelia Snyder.

The research team is a cross-functional department collaborating with the distribution, product, and engineering teams. Composed of professionals with substantial experience in the cryptoasset industry, our team places education at the core of our industrial research as we stand by free and publicly accessible content; and strongly believe information asymmetry contradicts the crypto ethos and philosophy. We provide data-driven, cutting-edge, unique insights into the crypto markets and macroeconomic factors likely to influence the state of this industry.

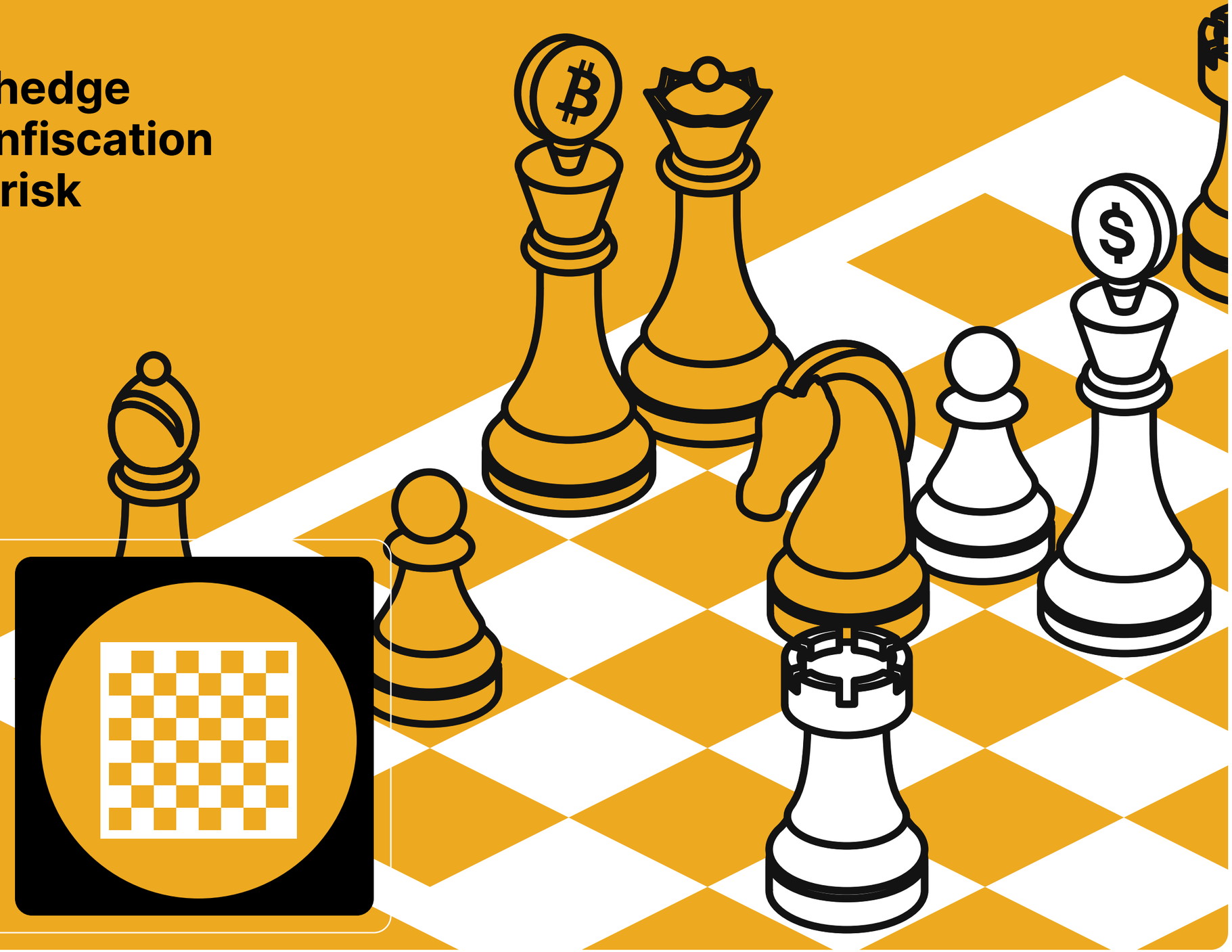
More than 10,000 investors read our research notes and reports on a weekly basis, ranging from private banks, asset managers, professional traders, hedge funds, tier-1 media outlets, and regulators.

Eliézer	Director of Research
Adrian	Senior Research Associate
Carlos	Research Analyst
Karim	Research Associate
Leena	Research Associate
Tom	Research Analyst



Bitcoin

The international hedge against wealth confiscation and counterparty risk



01 Bitcoin, the International and Censorship-Resistant Asset

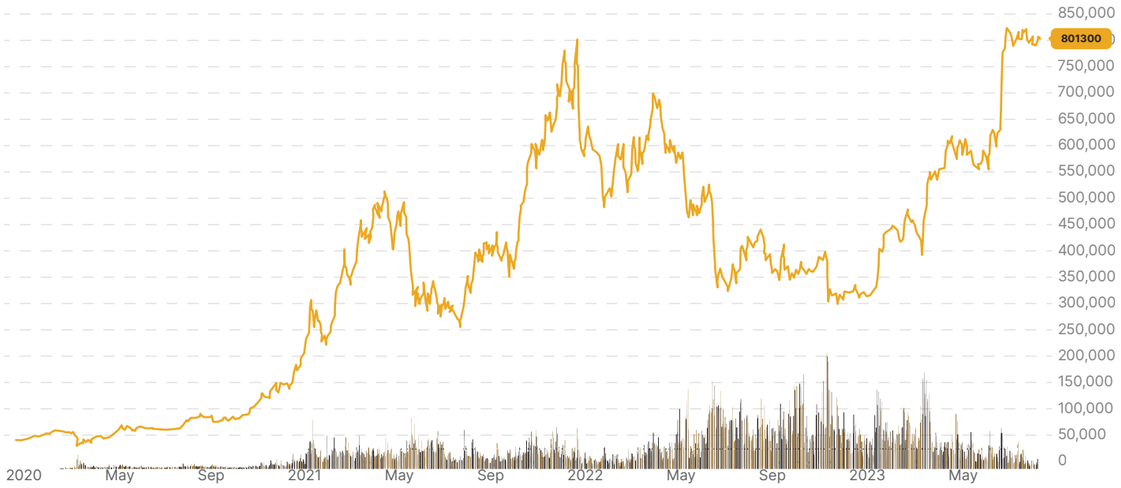
02 When Inflation Bites, Investors Turn to Crypto

03 Flight to Safety: Bitcoin or Stablecoins?

Bitcoin, the International and Censorship-Resistant Asset

Bitcoin was created in 2009 as a decentralized and internet-native asset with a monetary policy not controlled by centralized authorities in response to the Global Financial Crisis. Bitcoin (BTC) is increasingly recognized as an **emerging store of value** and a **potential hedge against currency debasement, wealth confiscation, and counterparty risk**. Impressively, there are now over 12 million wallets holding BTC, and even during challenging markets like in 2022, this number grew by over 20%. The recent banking crisis in the U.S. in March 2023 was a testament to Bitcoin's product-market fit, which led to a 40% rally.

Figure 1 – Bitcoin's price against the Turkish lira (TRY)



Sources: TradingView

When Inflation Bites, Investors Turn to Crypto

High inflation, geopolitical tension, and economic uncertainty drive demand for **Bitcoin as a neutral, international, and censorship-resistant asset**. Some nations, such as Venezuela or Turkey, have suffered the ravages of inflation for years due to economic mismanagement, social unrest, sanctions, and over-reliance on certain commodities like oil and gas. **Whatever the reason, many investors turn to crypto in a world with shifting geopolitical landscapes and sticky inflation**. While investors primarily measure Bitcoin's value in U.S. dollar (USD) terms, BTC has already approached or surpassed prior all-time highs in other fiat currencies including the Turkish lira (TRY), demonstrating its use case as a potential hedge against currency debasement. TRY has persistently eroded in the last three years, with a 30% drop against USD this year alone. The BTC/TRY price chart reveals a **significant uptick in trading volume between BTC and TRY**, reaching its highs in May 2023 and surpassing the previous peak in November 2021.

Table 1 – Fiat currencies inflation data

Country	Inflation (%)	Crypto adoption	% from BTC ATH
Venezuela	400%	10.30%	220%
Lebanon	253%	2.48%	350%
Argentina	98.60%	5.73%	30%
Turkey	50.60%	6.53%	4%
Ghana	45.40%	2.22%	-16%
Pakistan	27.10%	6.60%	-25%
Egypt	21.60%	3.04%	-10%
Ukraine	21.10%	10.57%	-56%
Nigeria	20.10%	5.93%	-14%
United States	4.50%	14.36%	-57%
Switzerland	2.40%	2.02%	-62%

Sources: imf.org, triple-a.io, google finance, tradingeconomics.com - data as of Aug-10,2023

Flight to Safety: Bitcoin or Stablecoins?

The U.S. dollar still stands as the world's reserve currency: 88% of the **$7.5 trillion**[1] traded daily in OTC FX markets in 2022 involve USD, per the BIS, and about half of the **$32 trillion**[2] global trade that year was invoiced in dollars, per the United Nations. In 2022, there was also a clear flight to safety, and the U.S. dollar was the preferred choice for most investors, evidenced by the DXY index, which spiked to highs not seen since 2001. Cryptoassets pegged to the dollar or stablecoins like Tether's USDT have gained traction as an **option against weakening local currencies** and for remittances. In 2022, **USDT settled $18.2 trillion in volume, surpassing payment giants like Mastercard and Visa**. Additionally, about **$530 billion**[3] **U.S. Treasuries and government bonds have been tokenized** in 2023, as of August 12.

Although the U.S. dollar and stablecoins may serve as safe havens, USD has experienced a significant decline in purchasing power over the past decades, see Figure 3. Both stablecoins and Bitcoin may act as a hedge against currency debasement, but there are key differences.

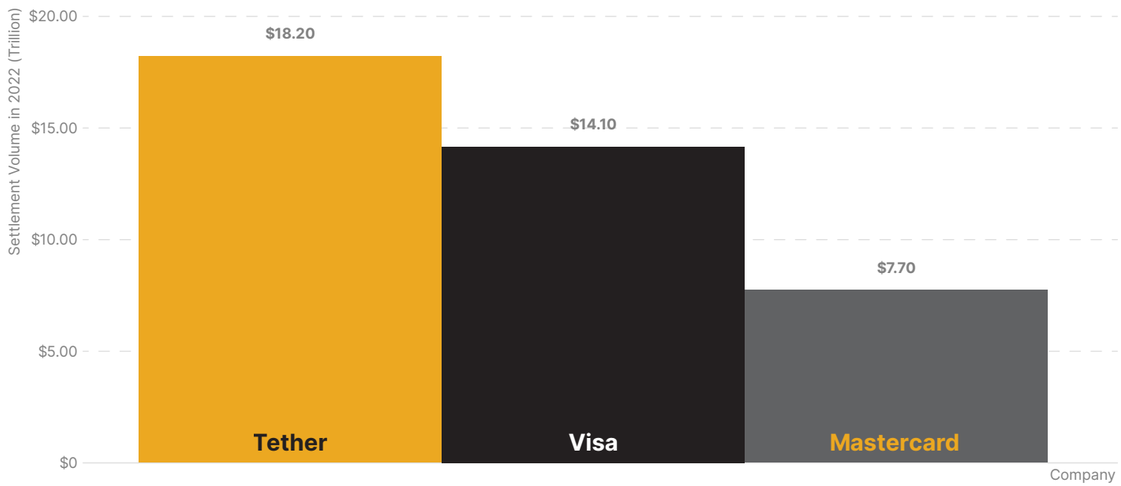
- The main advantages of Bitcoin compared to stablecoins include its **decentralization, self-custody, and censorship resistance**.

- Centralized stablecoin issuers like Tether (USDT) and Circle (USDC) maintain the right to freeze accounts, **exposing investors to counterparty risk**.

We will explore in the next section, the various use cases for stablecoins including cross-border payments.

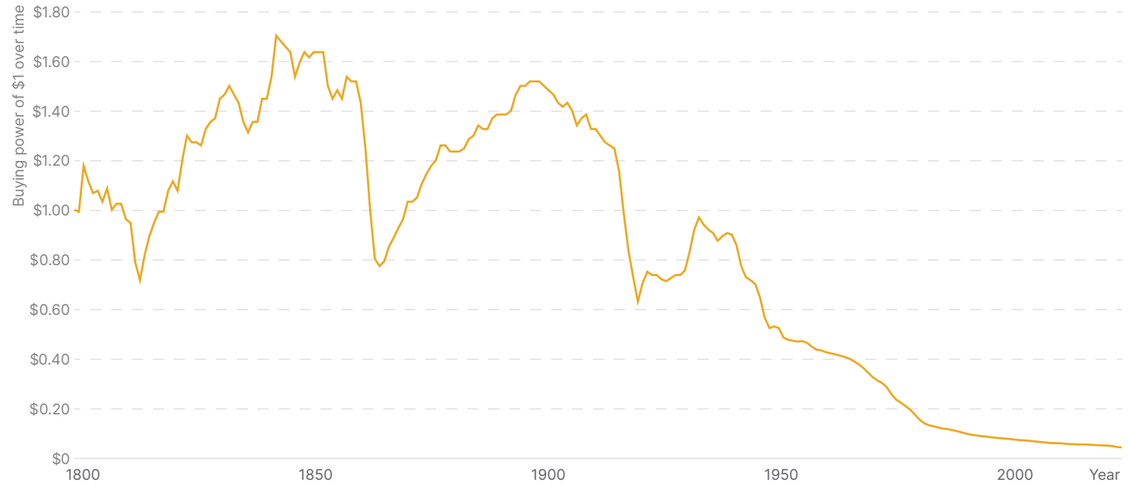
> "The main advantages of Bitcoin compared to stablecoins include its decentralization, self-custody, and censorship resistance."

Figure 2 – Settlements in 2022 vs. credit card networks



Source: Twitter - Tether

Figure 3 – Buying power of $1 over time, 1800-2023



Source: officialdata.org

USD Stablecoins

Near-instant and cheap cross-border payments with digital dollars



01 USD Stablecoins, Near-instant and Cheap Cross-Border Payments With Digital Dollars

02 If Crypto Is So Much Better Than Traditional Payment Methods, Why Is It Not Mainstream?

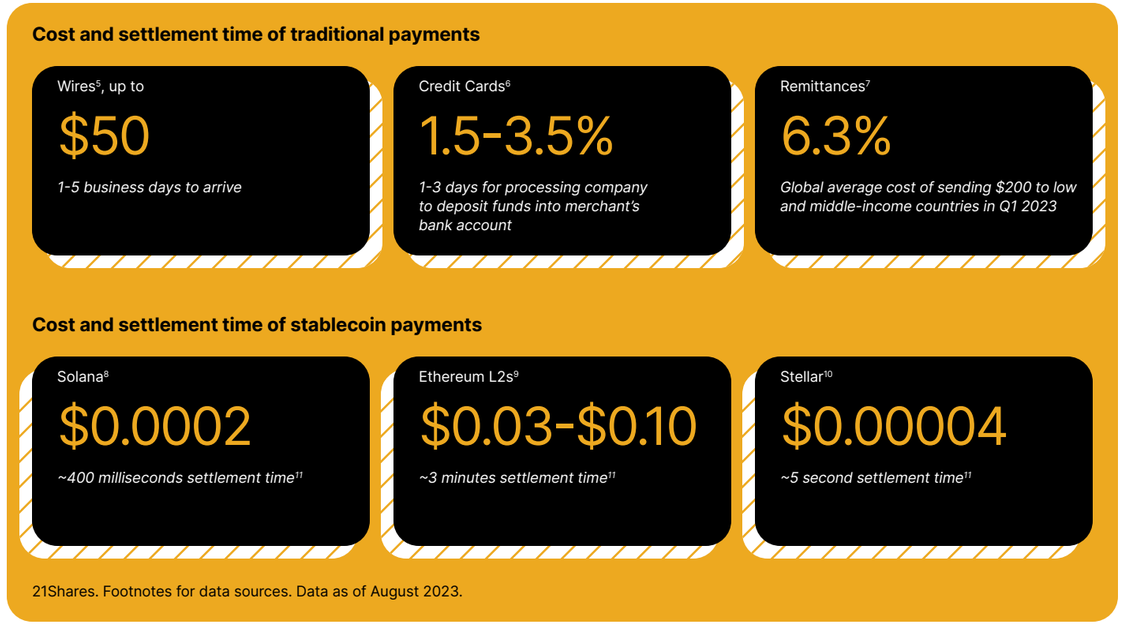


USD Stablecoins, Near-Instant and Cheap Cross-Border Payments With Digital Dollars

Crypto is the next logical evolution of money and the internet. Although the internet gave rise to a global and open network for information, our financial system is still a relic of the past, a siloed infrastructure imposing trillions of dollars of cost and delays on people and businesses. By contrast, **blockchain networks enable us to transfer value as easily as sending an email, globally, almost instantly, and cheaply**. Anyone with an internet connection can obtain a free wallet to store and transact with cryptoassets like Bitcoin or Ether, or USD stablecoins like USDT and USDC. Crypto is ripe to disrupt the **$37 trillion cross-border payment industry**[4] over the coming years, similar to how the instant flow of data over the internet disrupted legacy industries such as newspapers, publishers, and video rental stores.

Traditional payment systems are geographically fragmented, resulting in slow, expensive, and opaque fee structures for cross-border transfers. In an increasingly digital and globalized world, it's incomprehensible that people still worry about banking hour restrictions. Fintech companies emerged to address some of these issues over the past 15 years, but their digitization is still limited by legacy rails. On the other hand, blockchains use internet-native rails, allowing money to broadcast as fast and cost-efficiently as other forms of data. For instance, **sending USDC on Solana costs less than a cent, and the settlement time is less than a second**. Because it costs no more to send a $100 or $100 million transaction, crypto can provide value to anyone, from people sending remittances to loved ones abroad to large corporations paying overseas suppliers.

Figure 4 – Cost and settlement time of traditional vs. stablecoin (USDC) payments



21Shares. Footnotes for data sources. Data as of August 2023.

If Crypto Is So Much Better Than Traditional Payment Methods, Why Is It Not Mainstream?

In our view, two main factors are holding back the adoption of stablecoins for cross-border payments:

1. **Low internet penetration rate in emerging economies:** The global internet penetration rate sits barely above 60%, with some regions, such as South Asia and Sub-Saharan Africa, having penetration rates below 40%.
2. **Most digital wallets lack robust and user-friendly payment functionality:** People and businesses still have to rely on the traditional financial system, so it's essential to integrate crypto with existing financial software for seamless on and off-ramps.

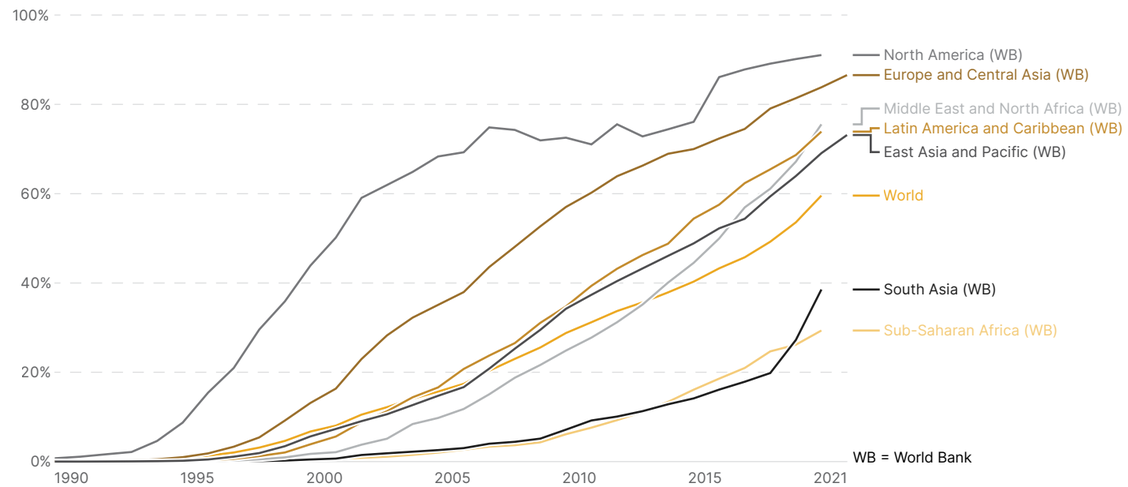
How can we overcome these limitations?

- **Developing regions are quickly adopting crypto:** Internet users are growing at an annual rate of 2.1%[12], while growth is much higher in developing economies. People already use crypto for cross-border payments where the internet is broadly accessible. For instance, Bitso, a crypto exchange, processed **$3.3 billion[13] in remittances** between the U.S. and Mexico (~79%[14] internet penetration rate) in 2022, the largest remittance corridor in the world with $52.6 billion of flows. In addition, Circle estimates[15] that **~70% of USDC adoption lies outside the U.S.**, with emerging economies standing out as some of the fastest-growing areas.

- **Seamless cash on and off-ramps:** About 2 billion workers[16] worldwide operate in the informal sector, with a high cash prevalence. Thanks to Stellar's "anchor network[17]," which includes a partnership with money transfer company MoneyGram, USDC on Stellar now has **over 81,000 on-ramp and over 320,000 off-ramp locations worldwide**.

- **Integrations with traditional payment methods:** Stablecoin use in merchant payments could grow widespread in the coming years as crypto integrates traditional payment methods. For example, digital payments giant **PayPal recently launched a USD stablecoin (PYUSD[18])**, citing a "shift toward digital currencies." In addition, Gnosis Pay[19] introduced a **self-custodial Visa debit card** that allows users to spend their crypto like they would spend cash in their bank accounts.

Figure 5 – Share of the population using the Internet

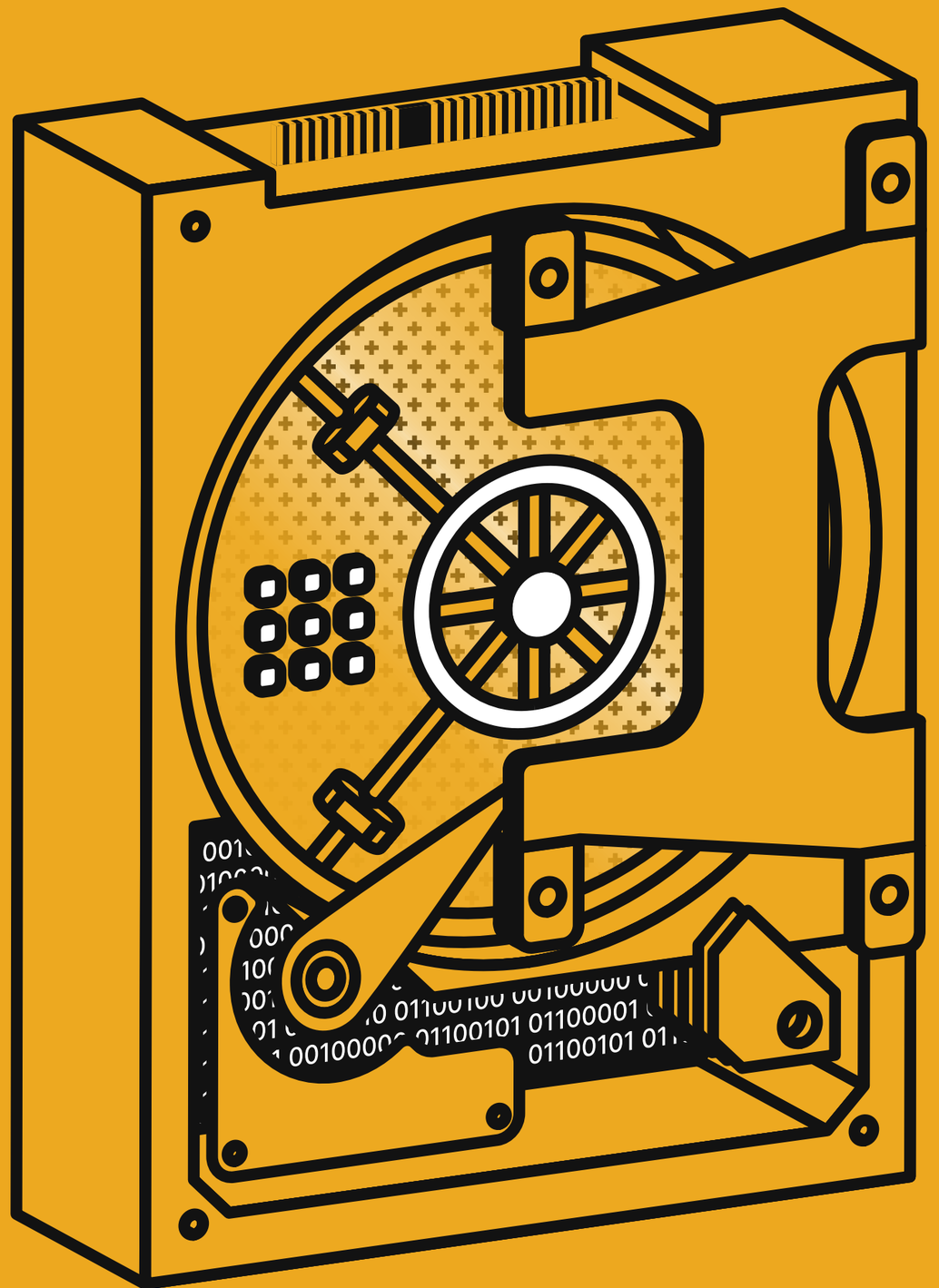


1. Internet user: An internet user is defined by the International Telecommunication Union as anyone who has accessed the internet from any location in the last three months. This can be from any type of device, including a computer, mobile phone, personal digital assistant, games machine, digital TV, and other technological devices.

Source: Our World in Data[20]

Data Storage

An immutable hard drive for preserving humanity's knowledge



01 Data Storage, An Immutable Hard Drive for Preserving Humanity's Knowledge

02 What Are the Current Limitations of Decentralized Storage Networks?



Data Storage, an Immutable Hard Drive for Preserving Humanity's Knowledge

The blockchain's distributed architecture form the basis of decentralized data storage protocols, which prevent **data loss and censorship**, enabling to store information perpetually. These protocols host critical data, such as records of **Russian violations in Ukraine and the censored content of Hong Kong's pro-democracy tabloid, Apple Daily**. With the volume of data stored projected to grow 150%[21] by 2025, these protocols provide a timely and vital solution resistant to government censorship and central points of failure. Compared to the **$230 billion[22] data storage market**, the decentralized storage (dStorage) industry is still nascent, valued at **$5.5 billion**, with significant room for growth. Unlike central data centers, decentralized storage protocols **distribute** data across a network of computers, leveraging their **unused** resources as global immutable hard drives. The setup helps maintain data integrity, safeguard against intrusion, and prevent tampering by **replicating** the ledger across individuals residing at geographically distributed locations.

Decentralized storage offers **cost advantages** by eliminating operational expenses tied to data centers, although pricing varies based on storage type and size. Centralization of data centers also exposes data to potential loss and theft. The **2023 Verizon Data Breach Investigation report[23]** documented over 32,000 security incidents in storage centers, with nearly 60% reporting personal data being exposed. Finally, Web 2 storage offerings are walled gardens, leading to complications when sharing data across different organizations. Hence, decentralized storage protocols solve the following problems:

- **Distribute data across computers to prevent total loss due to individual failures**.

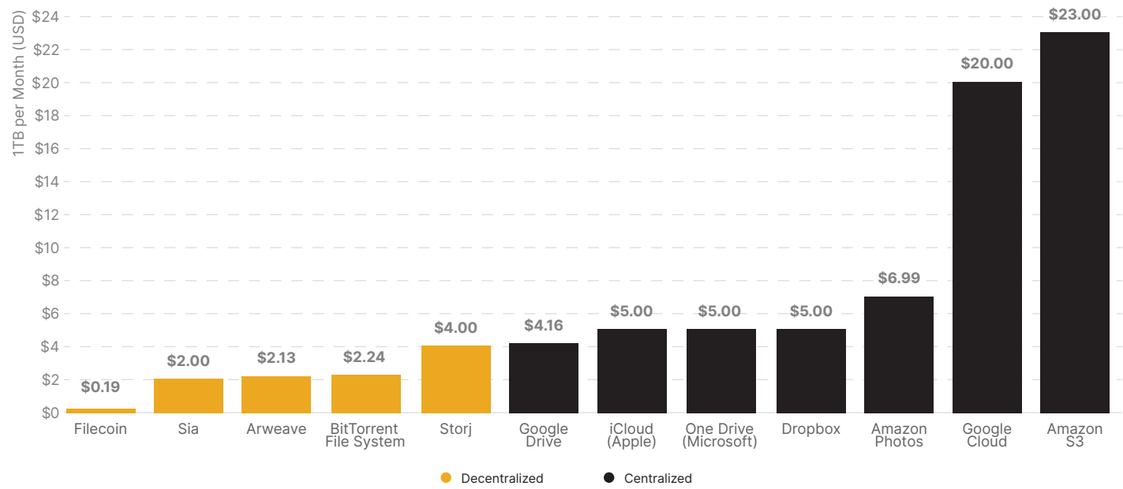
- **Replicate data to prevent single-entity control and offer perpetual accessibility**. For instance, Arweave stored records of Russian violations in Ukraine when Putin ordered social media to be shut down and Apple Daily's pro-democracy movement publication in Hong Kong, which the Chinese government censored.

- **Directly owning data, controlling its access, and retaining value through participation in data marketplaces**. For example, *Mirror* – a Medium-competitor hosted on the Arweave blockchain – rewards content creators by proving ownership of their work and reimbursing them in an automated manner.

- **Data interoperability**. Healthcare, general sciences companies, and other industries could benefit from sharing sensitive data without an intermediary. For instance, **37% of life sciences and environment-related companies use the Filecoin Plus program** to store various data that are challenging to share securely and efficiently, such as satellite imagery, DNA sequencing data, and neuroimaging datasets.

> "These protocols host critical data, such as records of Russian violations in Ukraine and the censored content of Hong Kong's pro-democracy tabloid, Apple Daily."

Figure 6 – Cost of decentralized storage vs. centralized storage in 2023
Storage provider pricing for 1TB per month (May 17, 2023)



Source: Coingecko[24]

What Are the Current Limitations of Decentralized Storage Networks?

- **Weak network effects** discourage users from migrating to unfamiliar services. Crucially, existing storage solutions **lack compatibility** with dStorage protocols, forcing reliance on manual data transfers and technical expertise.

- **Blockchain's architecture also leads to data fragmentation and network latency**, limiting dStorage's broad use and confining it to specific cases that don't require quick retrieval.

- **Both market leaders lack intuitive interfaces** that abstract away the complexity of uploading data, although that is now positively changing.

- Finally, **the inability to verify the authenticity of external data** means that users have to trust the data provider.

How can we overcome these limitations?

- **Decentralized storage protocols must integrate with existing solutions. For instance, Filecoin toolkits**[25] import data directly from browsers and existing Web 2 offerings like Amazon Web Services (AWS) and Microsoft Azure.

- As discussed in the previous section, **crypto needs robust payment integrations. Arweave** launched a service dubbed **Turbo**[26], allowing users to pay with credit cards when uploading data while converting to the required AR token in the backend.

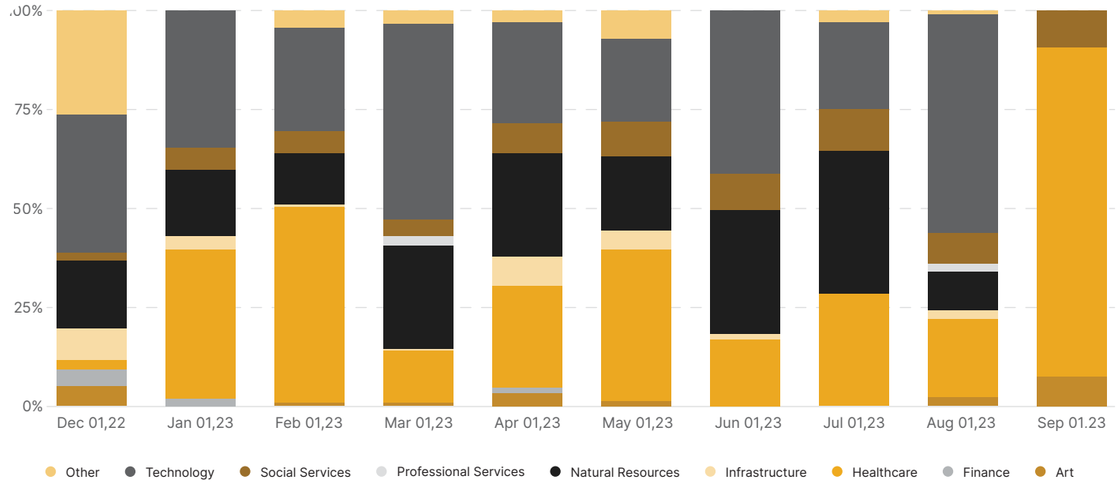
- **User-centric interfaces are improving by orders of magnitude**. For example, Arweave's "Google-like" ArDrive enables phones to upload data and **Filecoin Data Tools (FDT)**, simplifying the data onboarding process.

- **Innovative proving systems have the potential to verify the authenticity of external data**. For instance, *Space and Time Proof-of-SQL* corroborates the authenticity of data imported from the real world and ensures queries are accurate and unmodified.

- **Web 3 players will likely be the primary consumers of data storage.** Crypto-native apps will help drive decentralized storage adoption, particularly in smart contract platforms and NFT-based content, given their essential demand for continuous data availability. To illustrate, Solana and Lens Protocol depend on Arweave to store extensive data, alleviating network congestion.

"Crypto-native apps will help drive decentralized storage adoption, particularly in smart contract platforms and NFT-based content, given their essential demand for continuous data availability."

Figure 7 – Monthly client breakdown by industry in %



Source: Starboard.ventures[27]

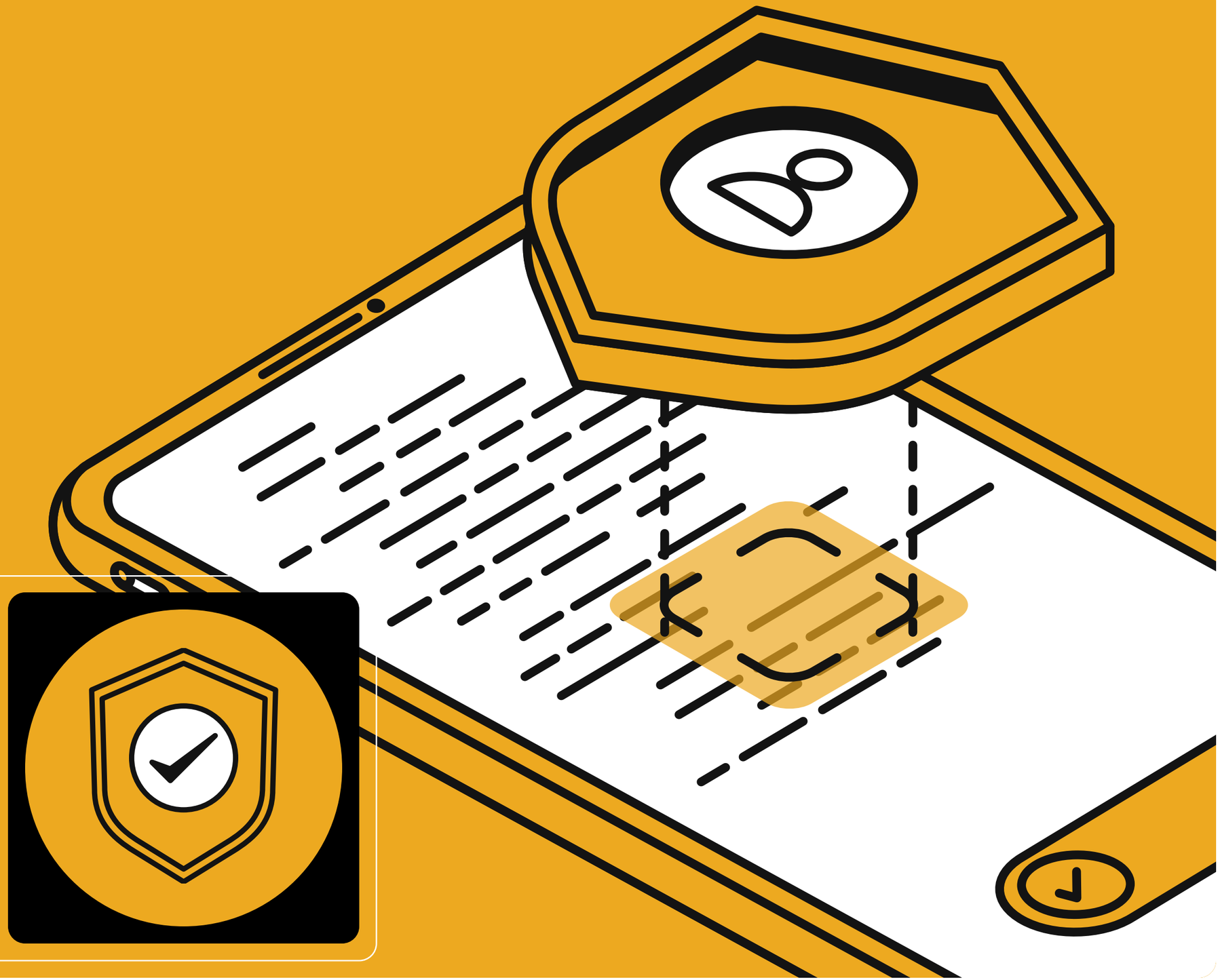
Data as of September 14, 2023

Digital Identity

Protecting online authenticity

01 How Can Blockchains Protect Our Identity?

02 Fighting the Next Generation of Identity Theft: Sybil Attacks



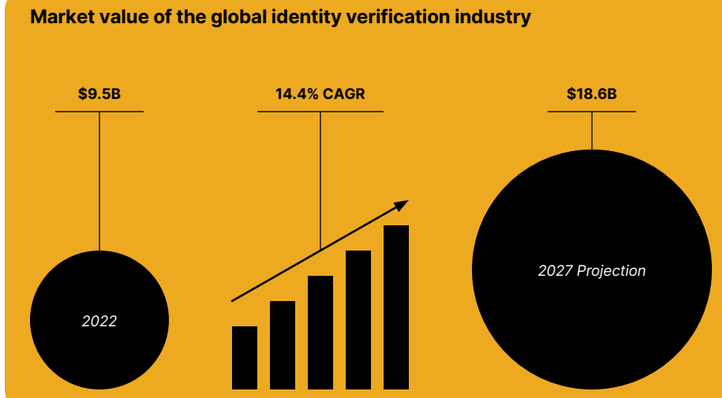
How Can Blockchains Protect Our Identity?

Blockchains can permanently safeguard identification documents (IDs), preventing damage, data breaches, and identity theft. **The latter alone cost the world $8.8 billion[28] in 2022**. Decentralized identity (also known as DID) solutions account for 0.07% of the crypto industry's market cap, but **over 4 million users are already leveraging the top decentralized IDs alone**. Although there are countless use cases for decentralized identity, one echoes the loudest. According to the World Economic Forum, about **850 million[29] people lack IDs worldwide** due to loss and damage or complicated registration regimes. People at risk of displacement would benefit the most from decentralized identity solutions. Nonetheless, as technology advances on both fronts of good and bad actors, anyone could be at risk of identity exploitation.

In 2017, a war broke out in Myanmar, leading to a population of over 700 thousand to flee their homes without packing their most valuable belongings, including official documents. As they settled in Bangladesh, the Rohingya refugees couldn't find a solution to their identity crisis besides blockchain technology. A Rohingya refugee, Mohamad Noor, launched **the Rohingya Project[30], a decentralized identity ecosystem** to empower their community with identity parameters to grant them access to healthcare, financial support, and other fundamental human rights. The Rohingya Project illustrates the main value proposition of decentralized identity solutions, **identifying and authenticating users and entities with more efficient security standards than their traditional counterparts**. Identity management uses attributes such as name, birth date, social security number, or email address to verify a person or entity to interact and transact with an online service.

> "The Rohingya Project illustrates the main value proposition of decentralized identity solutions, identifying and authenticating users and entities with more efficient security standards than their traditional counterparts."

Figure 8 – Market value of global identity verification industry vs. decentralized identity solutions



DID Solution	Market cap	Holders	IDs
Worldcoin	$577.5M	5,424	2,195,673
Ontology	$167.4M	1,458,537	1,633,700
Civic	$85.9M	66,018	350,000
Gitcoin	$59.4M	68,954	33,515
SelfKey	$27.3M	17,170	
Idena	$2M		1,571
ENS	$3.4M	65,033	735K

Sources: 21Shares, Emergen Research[31], Coingecko, Ontology explorer[32], Etherscan, Idena explorer[33]. Data as of August 7, 2023.

Fighting the Next Generation of Identity Theft: Sybil Attacks

By far, **the most important problem decentralized identity protocols must solve is Sybil attacks**, although further limitations are preventing the growth of DID solutions:

- Decentralized identity providers often struggle to **raise funds when looking to scale**. In The Hague, Tykn filed for bankruptcy in 2022 due to this obstacle. The award-winning decentralized identity solution had aimed to help ID the 474,000[34] stateless refugees in Europe.

- **Data exploitation concerns**. Collecting biometrics like iris scans performed by Worldcoin, for example, has raised questions and prompted regulators worldwide to launch preliminary investigations. Although Worldcoin claims that the data collected is preserved by a zero-knowledge (ZK) proof verification mechanism, the World Economic Forum warned against it, arguing that collecting this sensitive data from the outset is a dangerous tactic that decentralized identity protocols can live without.

How can we overcome these limitations?

- **Blockchains and decentralized identity protocols are ideally suited to fight Sybil attacks:**
    - **Ethereum** has made it harder for Sybil attackers to exploit its network by making running a node expensive, with a starting staking price of 32 ETH (around $60K).
    - **Gitcoin** has made their passports resistant to Sybil attacks by using a set of verification methods or "stamps," including **Idena's** DID solution.
    - **SelfKey** ID solution leverages artificial intelligence (AI) to detect fake, AI-generated digital identities and identity thieves.

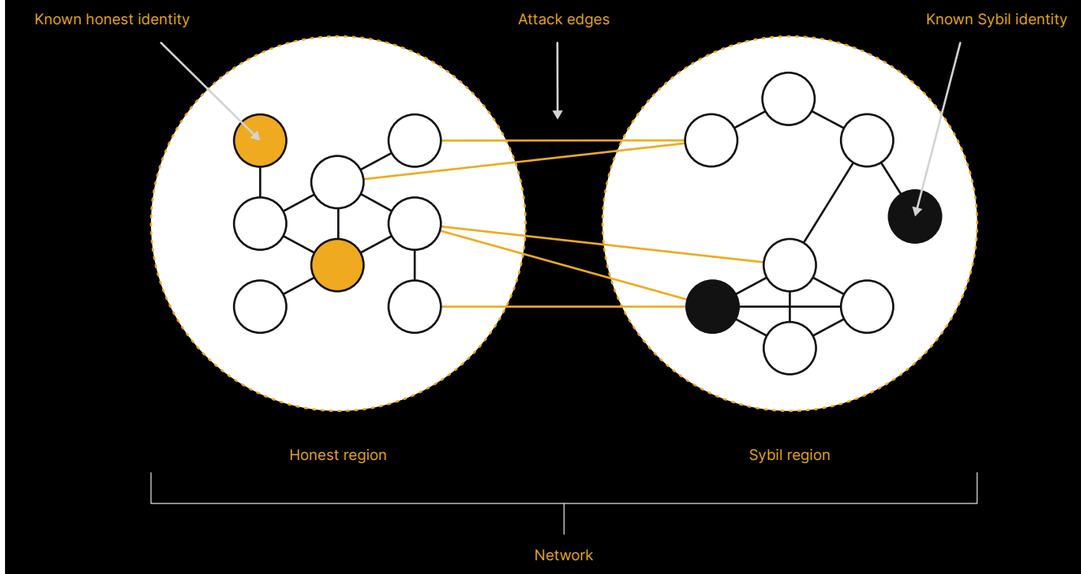
- With the deployment of AI in more aspects of our lives, there will be a higher demand for DID solutions to authenticate users. **Detecting misinformation generated by AI was valued at around $4 billion[35] in 2022**, according to findings by global research publisher HSRC. Fortunately, AI and machine learning are still underdeveloped, buying the DID industry more time to close all the loopholes and vulnerabilities these technologies can exploit.

- The DID industry **still needs recognition from transnational authorities like the United Nations**, which would lead to a more sustainable pace of adoption, as non-benefiting, neutral entities monitor more adequate data collection practices.

"With the deployment of AI in more aspects of our lives, there will be higher demand for DID solutions to authenticate users."

Figure 9 – What is a Sybil attack?

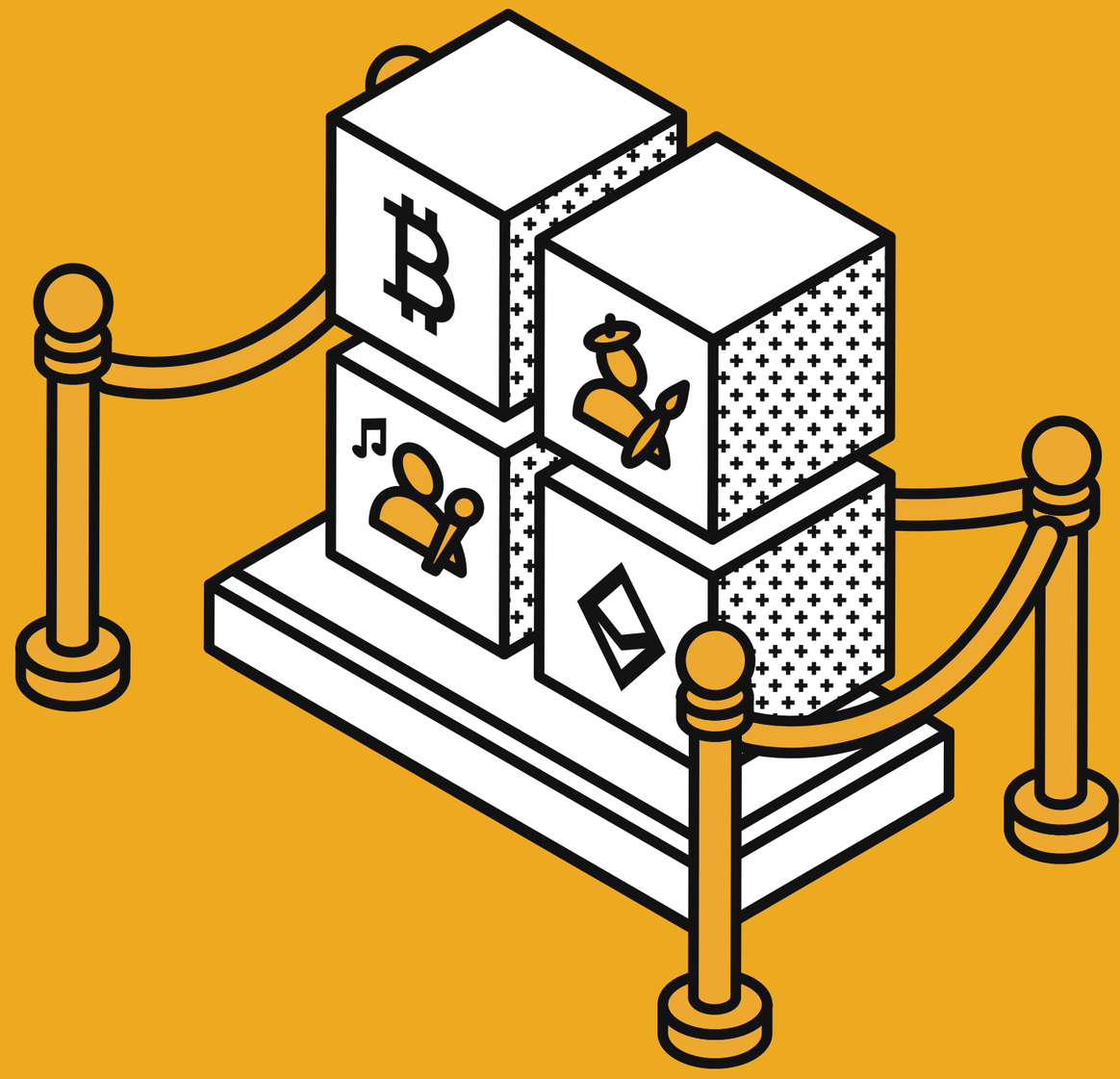
A Sybil attack is a cyber-attack that exploits a single node to generate many fake ones and undermine the authority of a peer-to-peer network, mainly targeting ones running on reputation-based consensus mechanisms. In 2014, a Sybil attacker exploited The Onion Router - an anonymous social-messaging platform - for several months, intending to deanonymize its users.



Source: 21Shares

NFTs

Democratizing revenue for artists and musicians



01 Decentralization Helps Artists and Musicians To Abandon Middlemen

02 What Is Hindering the Adoption of NFTs for Artist and Musicians?



Decentralization Helps Artists and Musicians To Abandon Middlemen

Traditionally, intermediaries take **88%** of the revenue generated by musicians' work, while artists have to pay dealers, galleries, or platforms a commission fee for the profit earned from their artwork. With the help of non-fungible tokens (NFTs) and smart contracts, artists and musicians can now retain **98%+** of the revenue generated by their work. NFTs are unique digital assets that can be used to represent ownership of art, music, and other creative works. When an artist or musician creates an NFT for their work, smart contracts can help to distribute royalties whenever the NFT is sold automatically. Notably, artists and musicians can now have a more sustainable and equitable source of revenue by capturing royalties in the secondary market, which was previously impossible.

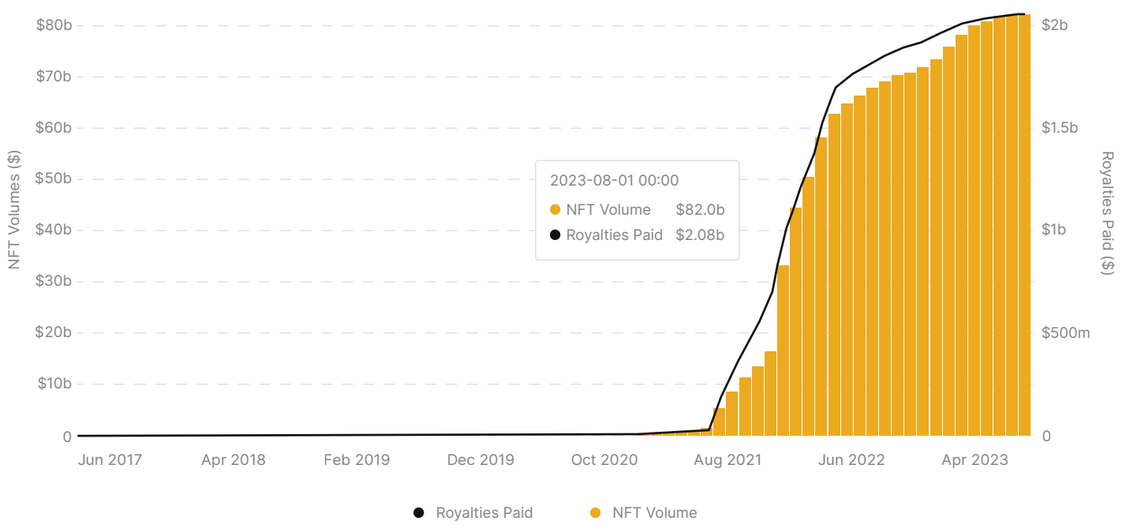
How can crypto help artists and musicians gain back the revenue of their work?

- **Near-zero charge to creators in primary sales:** Distribution channels take an important chunk of creators' work. In the music industry, a 2017 study by Citi[36] found that, in the U.S., artists only capture **12% of the revenue ($5.16B)**. Decentralized music protocols like Sound only charge a fixed fee of **0.0007 ETH (~$1.3)** per minting transaction on creators, and **100% of the revenue** from primary sales will go to artists.

- **Non-fungible nature helps digital artists to monetize revenue easier:** The critical challenges for monetizing digital art has been their fungibility and lack of ownership. Using NFTs, each piece of artwork has a unique identifier that represents ownership of it. Prominent auction houses Sotheby's[37] and Christie's[38] have partnered with digital artists Pak and Beeple, who have sold artwork for **$17 million** and **$69 million**, respectively.

- **Royalties, a sustainable source of revenue:** Creators usually do not get paid in any secondary market sale, missing out on a significant amount of income. Web 3 creators have received over **$2B[39] in royalties via smart contracts**. Large brands like Gucci, Adidas, Nike, and musicians like Snoop Dogg, Chainsmokers, and Nas have already been leveraging NFTs and generating a decent amount of revenue. NFTs allow artists to have a more sustainable source of income and be adequately compensated.

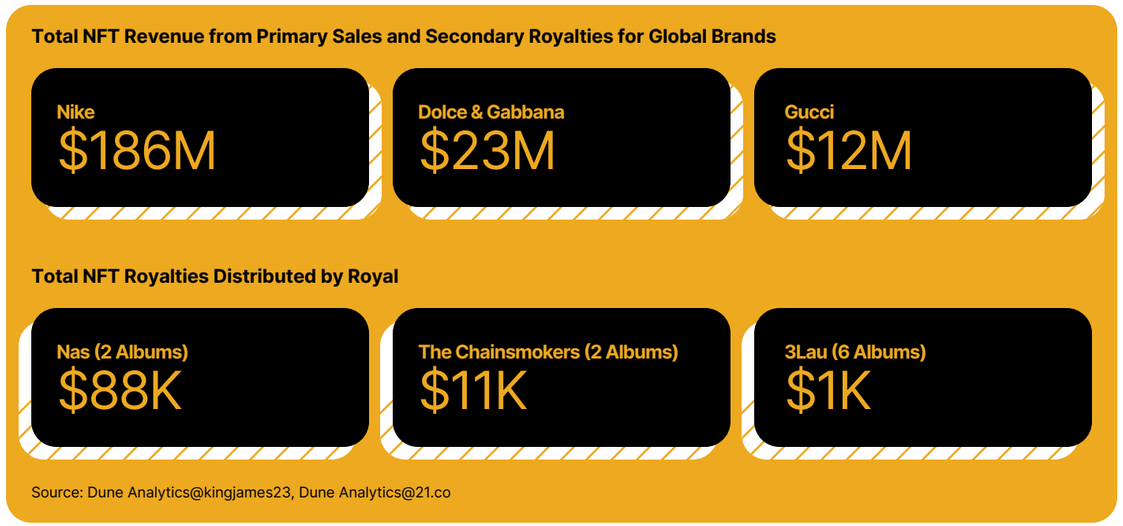
Figure 10 – Cumulative NFT secondary volume and royalties paid



Source: Dune Analytics@21.co

"With the help of non-fungible tokens (NFTs) and smart contracts, artists and musicians can now retain 98%+ of the revenue generated by their work."

Figure 11 – Revenue generated by global brands and artists via NFT sales



Source: Dune Analytics@kingjames23, Dune Analytics@21.co

What Is Hindering the Adoption of NFTs for Artist and Musicians?

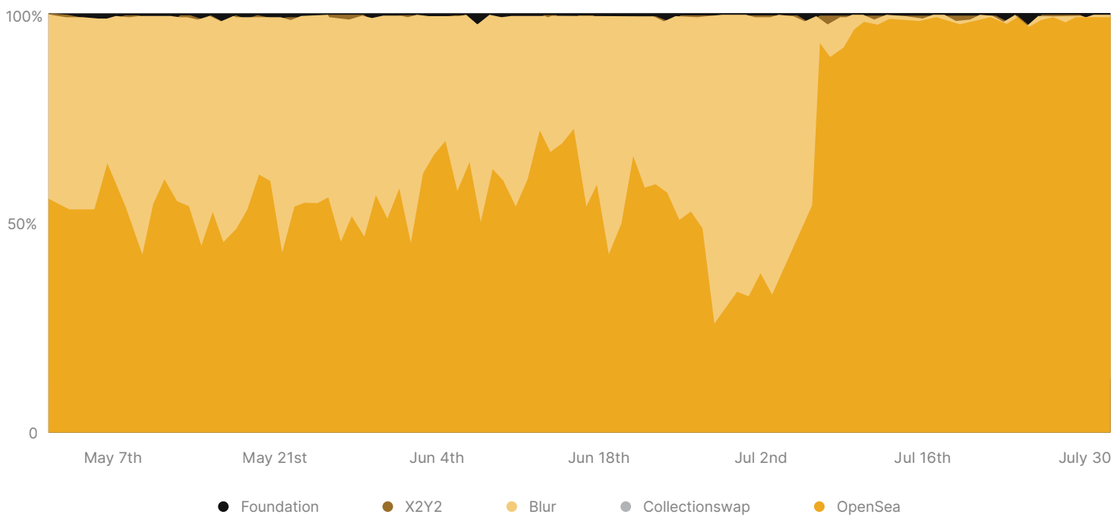
In our view, these are the key challenges limiting the adoption of NFTs for artists and musicians:

1. **Technical complexity of NFTs:** To launch an artwork or album as an NFT, creators must understand crypto and blockchain fundamentals. From learning how to use a wallet to off-ramping royalties earnings back to fiat currency, creators might be reluctant to spend extra time learning these things, instead of focusing on their creative work.
2. **Unenforceable on-chain royalties:** NFTs are usually implemented via the ERC-721 token standard. However, this standard does not enforce mandatory royalties. As a result, some NFT marketplaces like Sudoswap, LooksRare, X2Y2, and Blur used optional royalties to provide more competitive pricing for buyers. Currently, OpenSea is responsible for **99.7%**[40] of royalty fees despite Blur being the largest marketplace by traded volume.
3. **Exclusive deals:** Exclusive deals with streaming platforms like Apple Music and Spotify would prohibit artists from launching their albums on NFT marketplaces.
4. **Lack of mentorship:** Although intermediaries take a massive cut from artists and musicians' revenue, they do provide additional value, such as mentorship, financial management, and brand marketing, intangibles that a decentralized platform is not able to deliver at the moment.

How can we overcome these limitations?

1. **Education and better user experience:** Education is paramount to onboarding more users who understand the benefits of blockchain and NFTs. In the meantime, NFT marketplaces and music platforms could improve the user experience to allow creators to integrate their artwork seamlessly.
2. **ERC-2981, a new standard for royalties implementation:** ERC-2981 aims to add an additional layer for existing NFT token standards to have a standardised way to catalog the details of the royalties. Even though ERC-2981 can't enforce mandatory royalties, it helps marketplaces easily integrate royalty information without requiring creators to fill in the details on every marketplace. It set the global royalty payment standard and enabled support for royalty integration across marketplaces. Blue-chip NFT collections have already been integrated with the new token standard, such as Moonbirds, Checks, and Azuki.

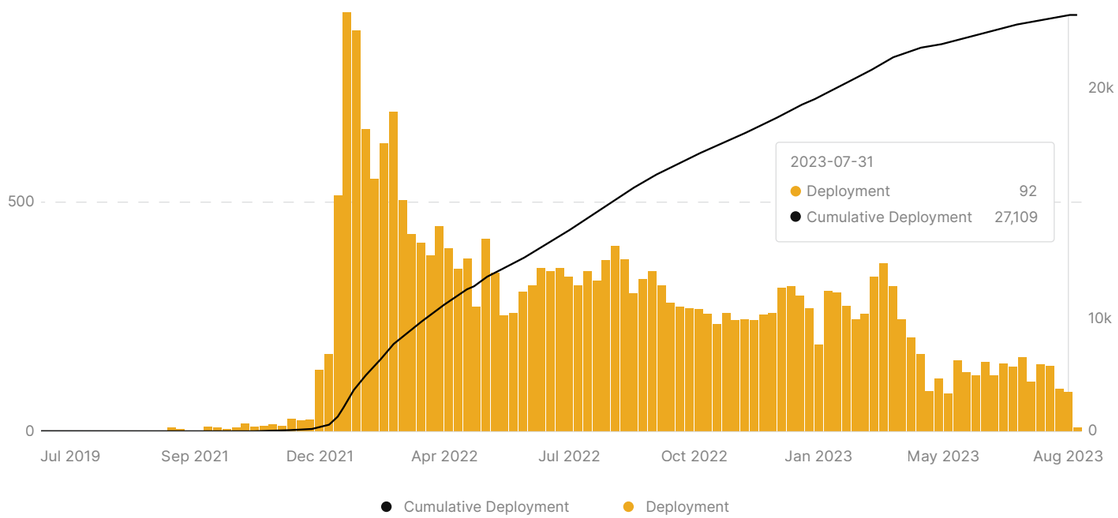
Figure 12 – Share of royalty fees among NFT marketplaces
Fees denominated in U.S. dollars



Source: Dune Analytics@hildobby

> "Education is paramount to onboarding more users who understand the benefits of blockchain and NFTs."

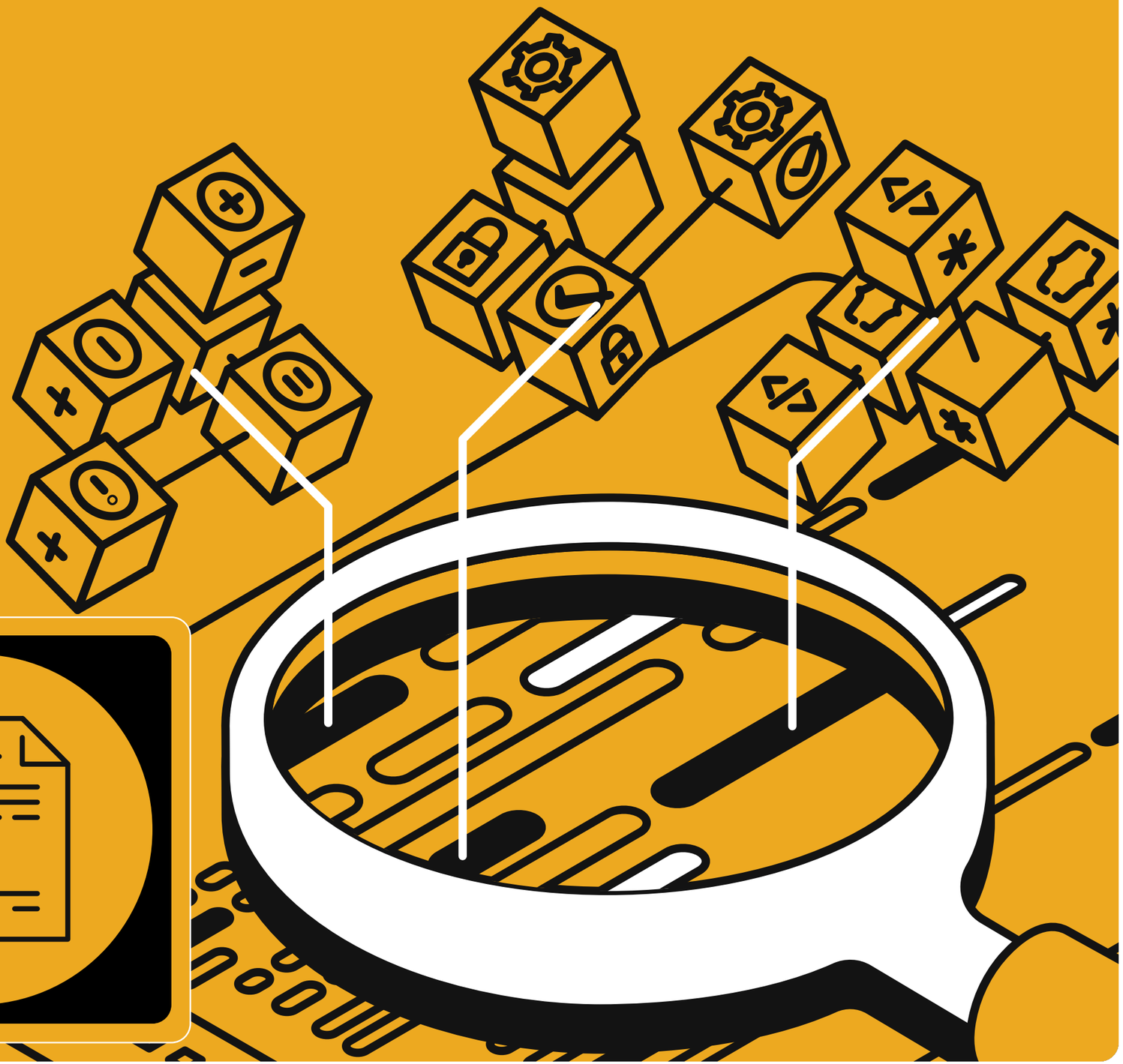
Figure 13 – Deployment of ERC-2981 NFT standard



Source: Dune Analytics@21.co and @ilemi

Forensic Science

Blockchain-as-a-service to combat illicit activities



01 Blockchains Are Inherently Transparent: Misconceptions About Illicit Activity in Crypto

02 Transparency As a Double-Edged Sword

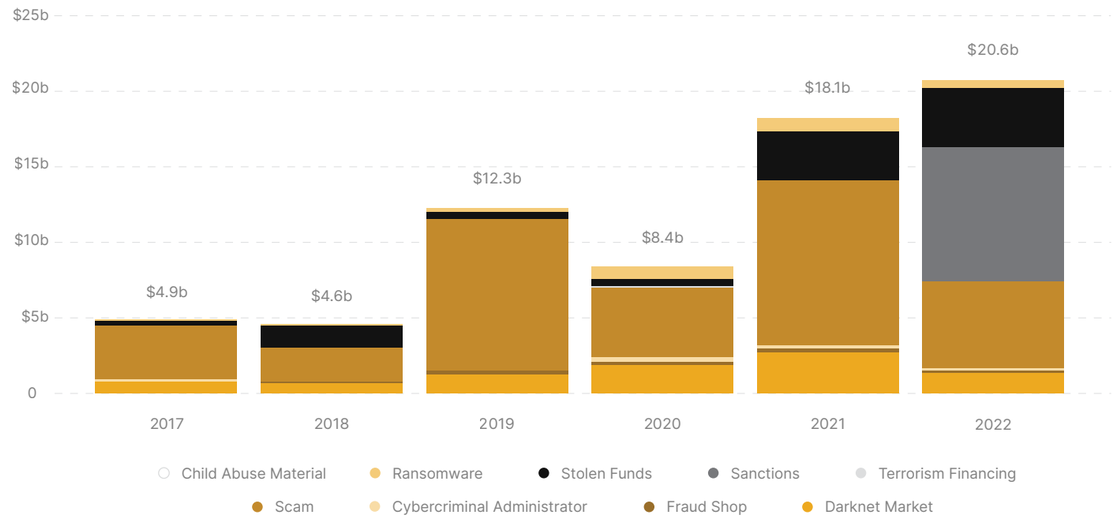


Blockchains Are Inherently Transparent: Misconceptions About Illicit Activity in Crypto

An early misconception about crypto that has been demystified over the years is that it can be used to obfuscate transactions easily and, thus, that it's regularly used for criminal purposes. According to Chainalysis, **illicit activity constituted less than 0.30% of crypto's total volume in 2022**. While it's true that there are a handful of protocols focused on privacy-enhancing features, Bitcoin, Ethereum, and the majority of cryptoassets today rely on a public blockchain that allows every transaction to be traced to a degree that isn't possible in traditional finance.

Illicit crypto volume reached an all-time high of at least **$20.6 billion** in 2023 amid a surge in sanctions designations and hacking, according to Chainalysis' 2023 Crypto Crime Trends[41] report. However, **43% of that volume came from activity associated with sanctioned entities** (e.g., OFAC sanctions[42] of Hydra, Tornado Cash, and Garantex), while every other illicit activity went down, with the exception of stolen funds. Furthermore, **only 0.24% of crypto's total volume was associated with illegal transactions**. In recent years, law enforcement has been able to trace stolen funds thanks to the transparency of the blockchain. For instance, the **U.S. Government has seized at least 215,000 BTC** since 2020, originating from Silk Road-related crimes and the 2016 Bitfinex hack. What's even more impressive, **our research team was able to track the holdings of the U.S. government in real time** on a data platform called Dune Analytics, leveraging that same transparency.

Figure 14 - Total crypto volume received by illicit addresses (2017-2022)



Source: Chainalysis

Transparency Is a Double-Edged Sword

Although the inherent transparency of blockchains solves many of the flaws of opaque and trust-based centralized systems, it also raises new challenges:

1. **Regulatory compliance risks:** Anyone with an internet-connected device can access crypto protocols by default, which introduces compliance risks like interacting with sanctioned entities or violating Anti-Money Laundering (AML) and Know Your Customer (KYC) guidelines.

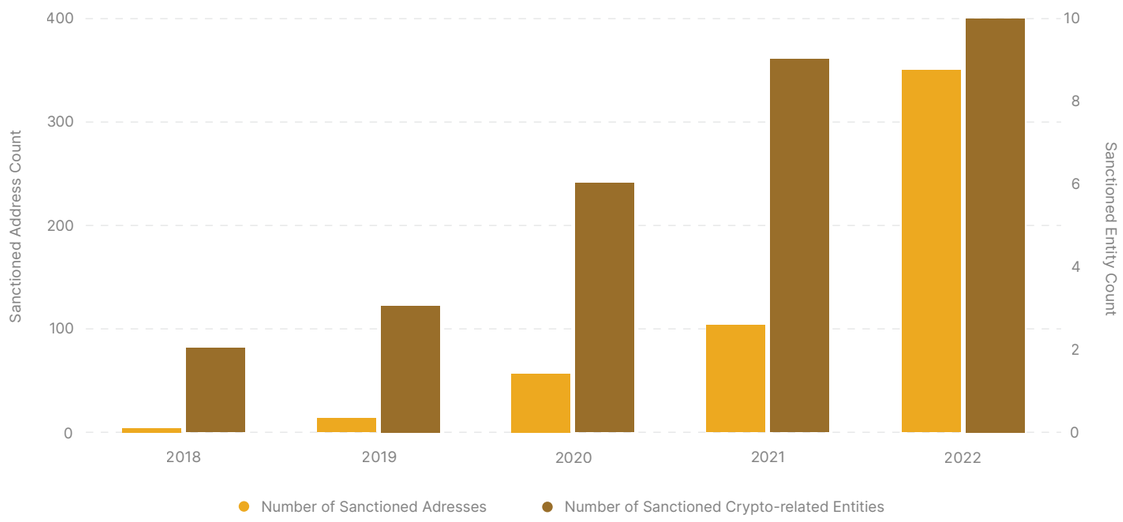
2. **Privacy concerns:** The transparency of blockchains raises surveillance concerns for individuals and prevents traditional institutions from adopting the technology. Today, companies with assets on-chain run the risk of malicious actors front-running or coordinating to target their liquidation levels in a given position.

How can we overcome these limitations?

- **Cryptoasset compliance software:** The Office of Foreign Assets Control (OFAC) of the U.S. Department of the Treasury sanctioned ten entities in 2022, with more than **300 wallet addresses** mentioned in the sanctions. The risk of having one of these addresses as a counterparty can be mitigated on-chain with the help of **forensic products such as KYT[43] (Know Your Transaction)** by Chainalysis, an API that helps organizations stay compliant with local and global regulations.

- **Privacy-enhancing protocols must balance preserving individual privacy while still complying with regulations:** For instance, the decentralized mixing service Tornado Cash was sanctioned in August 2022 by OFAC for "facilitating money laundering" related to funds stolen by North Korean hackers. Despite the complexity of the challenge, many protocols are proposing viable solutions that will launch gradually over the coming months. For example, Aztec revealed its "hybrid zk-rollup[44]," which will enable **private smart contract execution**, allowing users to protect their data and on-chain activity with programmable anonymity.

Figure 15 - Sanctioned crypto-related entities and number of sanctioned-related addresses by year added (2018-2022)



Source: Chainalysis

Conclusion

Crypto has grown by over 90,000% in market value over the past decade. Now that we have walked you through crypto's most pertinent use cases worldwide, it is important to note that these are not all. Some other groundbreaking use cases are still trying to find solutions for systemic inefficiencies in the traditional world. Given the magnitude of developments underway to overcome the infrastructure and regulatory obstacles, it will take some time for these solutions to morph into reality. That said, stay tuned for another edition of the State of Crypto, deep-diving into other use cases disrupting different segments of the world.

Footnotes

01. https://www.bis.org/statistics/rpfx22_fx.htm
02. https://unctad.org/system/files/official-document/ditcinf2023d1_en.pdf
03. https://app.rwa.xyz/treasuries
04. https://www.juniperresearch.com/pressreleases/cross-border-b2b-payments-to-surpass-$40tn
05. Bank Rate, https://www.bankrate.com/banking/wire-transfer-fees/
06. Forbes https://www.forbes.com/advisor/business/credit-card-processing-fees/ & Clearly Payments https://www.clearlypayments.com/blog/how-long-does-it-take-to-get-funds-in-credit-card-processing/#:~:text=In%20general%2C%20credit%20card%20processing,with%20a%20merchant%20account%20provider.
07. World Bank, https://remittanceprices.worldbank.org/sites/default/files/rpw_main_report_and_annex_q123_final.pdf
08. Solscan, https://beta-analysis.solscan.io/public/dashboard/06d689e1-dcd7-4175-a16a-efc074ad5ce2
09. Layer 2s or scaling solutions massively reduce transaction fees, making Ethereum more accessible for all. We expect this is where the majority of people and business will transact, which is why we focus on the cost of Ethereum L2s. https://l2fees.info/
10. Messari, https://messari.io/asset/stellar/metrics/fees-revenue
11. Circle, https://developers.circle.com/developer/docs/confirmations
12. https://datareportal.com/global-digital-overview#:~:text=The%20total%20number%20of%20internet,higher%20in%20many%20developing%20economies.
13. https://www.crowdfundinsider.com/2023/03/204447-digital-asset-firm-bitso-reportedly-facilitated-3-3b-in-institutional-crypto-transfers-between-mexico-and-us-last-year/
14. https://www.statista.com/statistics/227082/countries-with-the-highest-internet-penetration-rate/
15. https://twitter.com/jerallaire/status/1688672056898928640?s=20
16. https://www.imf.org/en/News/Articles/2021/07/28/na-072821-five-things-to-know-about-the-informal-economy#:~:text=The%20International%20Labor%20Organization%20estimates,%2D%2Dat%20least%20part%20time.
17. https://stellar.org/blog/bigger-is-better-why-stellar-is-the-leader-in-cash-to-crypto-on-and-off-ramps
18. https://www.paypal.com/us/digital-wallet/manage-money/crypto/pyusd
19. https://gnosispay.com/#how-it-works
20. https://ourworldindata.org/grapher/share-of-individuals-using-the-internet?time=1960..latest&facet=none
21. https://www.statista.com/statistics/871513/worldwide-data-created/
22. https://www.globenewswire.com/news-release/2023/02/28/2616751/0/en/With-17-8-CAGR-Data-Storage-Market-Size-Worth-USD-777-98-Billion-by-2030.html#:~:text=28%2C%202023%20(GLOBE%20NEWSWIRE),during%20the%202023%2D2030%20period.
23. https://www.verizon.com/business/resources/T6ab/reports/2023-data-breach-investigations-report-dbir.pdf
24. https://www.coingecko.com/research/publications/centralized-decentralized-storage-cost
25. https://dataonboarding.filecoin.io/
26. https://www.businesswire.com/news/home/20230719165456/en/ArDrive-Unveils-Turbo-a-Blockchain-Powered-Upload-Service-Redefining-Permanent-Data-Storage
27. https://dashboard.starboard.ventures/market-deals#datacap-applicants
28. https://www.ftc.gov/news-events/news/press-releases/2023/02/new-ftc-data-show-consumers-reported-losing-nearly-88-billion-scams-2022
29. https://www3.weforum.org/docs/WEF_Reimagining_Digital_ID_2023.pdf
30. https://www.theguardian.com/world/2018/aug/21/rohingya-turn-to-blockchain-to-solve-identity-crisis
31. https://www.emergenresearch.com/industry-report/identity-verification-market#:~:text=The%20global%20identity%20verification%20market%20size%20was%20USD%209.50%20Billion,the%20global%20identity%20verification%20market.
32. Ontology explorer https://explorer.ont.io/
33. Idena explorer https://scan.idena.io/
34. https://reporting.unhcr.org/operational/regions/europe
35. https://hsrc.biz/reports/counter-deepfake-counter-fakenews-market/
36. https://ir.citi.com/NhxmHW7xb0tkWiqOOG0NuPDM3pVGJpVzXMw7n%2BZg4AfFFX%2BeFqDYNfND%2B0hUxxXA
37. https://www.sothebys.com/en/articles/17-million-realized-in-sothebys-first-nft-sale-with-digital-creator-pak
38. https://onlineonly.christies.com/s/first-open-beeple/beeple-b-1981-1/112924
39. https://dune.com/21co/ethereum-key-metrics
40. https://dune.com/hildobby/nft-fees
41. https://go.chainalysis.com/2023-crypto-crime-report.html
42. https://blog.chainalysis.com/reports/how-2022-crypto-sanction-designations-affected-crypto-crime/
43. https://www.chainalysis.com/chainalysis-kyt/
44. https://medium.com/aztec-protocol/aztec-the-hybrid-zkrollup-a90a197bf22e

Contact

Research
research@21shares.com

Disclaimer

This document is not an offer to sell or a solicitation of an offer to buy or subscribe for securities of 21Shares AG. Neither this document nor anything contained herein shall form the basis of, or be relied upon in connection with, any offer or commitment whatsoever in any jurisdiction. This document and the information contained herein are not for distribution in or into (directly or indirectly) the United States, Canada, Australia or Japan or any other jurisdiction in which the distribution or release would be unlawful.

This document does not constitute an offer of securities for sale in or into the United States, Canada, Australia or Japan. The securities of 21Shares AG to which these materials relate have not been and will not be registered under the United States Securities Act of 1933, as amended (the "Securities Act"), and may not be offered or sold in the United States absent registration or an applicable exemption from, or in a transaction not subject to, the registration requirements of the Securities Act. There will not be a public offering of securities in the United States.

This document is only being distributed to and is only directed at: (i) to investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (the "Order"); or (ii) high net worth entities, and other persons to whom it may lawfully be communicated, falling within Article 49(2)(a) to (d) of the Order (all such persons together being referred to as "relevant persons"); or (iv) persons who fall within Article 43(2) of the Order, including existing members and creditors of the Company or (v) any other persons to whom this document can be lawfully distributed in circumstances where section 21(1) of the FSMA does not apply. The securities are only available to, and any invitation, offer or agreement to subscribe, purchase or otherwise acquire such securities will be engaged in only with, relevant persons. Any person who is not a relevant person should not act or rely on this document or any of its contents.

In any EEA Member State (other than the Austria, Belgium, Croatia, Denmark, Finland, France, Germany, Great Britain, Hungary, Ireland, Italy, Liechtenstein, Luxembourg, Malta, The Netherlands, Norway, Poland, Romania, Slovakia, Spain and Sweden) that has implemented the Prospectus Regulation (EU) 2017/1129, together with any applicable implementing measures in any Member State, the "Prospectus Regulation") this communication is only addressed to and is only directed at qualified investors in that Member State within the meaning of the Prospectus Regulation.

Exclusively for potential investors in Austria, Belgium, Croatia, Denmark, Finland, France, Germany, Great Britain, Hungary, Ireland, Italy, Liechtenstein, Luxembourg, Malta, The

Netherlands, Norway, Poland, Romania, Slovakia, Spain and Sweden the 2021 Base Prospectus (EU) is made available on the Issuer's website under www.21Shares.com.

The approval of the 2021 Base Prospectus (EU) should not be understood as an endorsement by the SFSA of the securities offered or admitted to trading on a regulated market. Eligible potential investors should read the 2021 Base Prospectus (EU) and the relevant Final Terms before making an investment decision in order to understand the potential risks associated with the decision to invest in the securities. You are about to purchase a product that is not simple and may be difficult to understand.

This document constitutes advertisement within the meaning of the Swiss Financial Services Act (the "FinSA") and not a prospectus. In accordance with article 109 of the Swiss Financial Services Ordinance, the Base Prospectus dated 12 November 2021, as supplemented from time to time and the final terms for any product issued have been prepared in compliance with articles 652a and 1156 of the Swiss Code of Obligations, as such articles were in effect immediately prior to the entry into effect of the FinSA, and the Listing Rules of the SIX Swiss Exchange in their version in force as of January 1, 2020. Consequently, the Prospectus has not been and will not be reviewed or approved by a Swiss review body pursuant to article 51 of the FinSA, and does not comply with the disclosure requirements applicable to a prospectus approved by such a review body under the FinSA.